Gravity as Information Recovery: The Gravitational Refractive Index is the Petz Fidelity Bound

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Three quantities from three fields turn out to be the same number. From general relativity: the coordinate speed of light $c_{\rm eff}/c = e^{\Phi/c^2}$. From thermodynamics: the entropy production cost $e^{-\Sigma_{\rm grav}/2}$ with $\Sigma_{\rm grav} = 2|\Phi|/c^2$. From quantum information: the Petz recovery fidelity bound $F_{\rm bound} \geq e^{-\Sigma/2}$. The identification $c_{\rm eff}/c = e^{-\Sigma_{\rm grav}/2} = F_{\rm bound}$ states that light slowing in a gravitational field *is* information loss, and the gravitational refractive index *is* the inverse recovery fidelity. Three independent routes—modular flow, gravitational Landauer principle, and quantum channel analysis—converge on $\Sigma_{\rm grav} = -\ln(-g_{00})$. The identity $\sqrt{-g_{00}} = e^{-\Sigma_{\rm grav}/2}$ yields the exponential metric $g_{00} = -e^{-r_{\rm s}/r}$, which has no event horizon ($\tau < 1$ everywhere) and agrees with Schwarzschild through second post-Newtonian order. Strong-field deviations—4.6% larger shadow, $\sim 4\%$ shifted quasinormal modes, and gravitational-wave echoes at $\Delta t \approx 4\, r_{\rm s}/c$—are within reach of current observatories.

INTRODUCTION

In 1911, Einstein observed that the speed of light varies in a gravitational field [?]. Dicke later showed that general relativity (GR) can be exactly reformulated as flat-spacetime optics with a gravitational refractive index [?]: in a static field with Newtonian potential $\Phi < 0$,

$$\frac{c_{\rm eff}}{c} = e^{\Phi/c^2}\,. \tag{1}$$

Meanwhile, quantum information theory has established that any quantum channel $\mathcal{N}$ with entropy production $\Sigma \equiv D(\rho\|\sigma) - D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma))$ admits a universal recovery bound [? ?]:

$$F\big(\rho,\ \mathcal{R}_\sigma \circ \mathcal{N}(\rho)\big)\ \geq\ e^{-\Sigma/2}\,, \tag{2}$$

where $\mathcal{R}_\sigma$ is the (rotated) Petz recovery map [?] and F is the Uhlmann fidelity. The Petz map is the *unique* retrodiction functor satisfying Bayesian consistency [?]—the quantum generalization of Bayes' rule. In a companion paper [?], we showed that the temporal asymmetry parameter $\tau = 1 - F \in [0, 1]$ provides an operational measure of the arrow of time, with $\tau = 0$ for closed systems and $\tau \leq 1 - e^{-\Sigma/2}$ in general.

Here we connect these two results. Equations (**??**) and (**??**) share the same exponential structure. This is not a coincidence: they are the same physical statement in different languages.

THE TRIPLE IDENTIFICATION

We define the dimensionless gravitational entropy production as

$$\Sigma_{\rm grav}\ \equiv\ \frac{2|\Phi|}{c^2}\ =\ \frac{r_{\rm s}}{r}\,, \tag{3}$$

where $r_{\rm s} = 2GM/c^2$ is the Schwarzschild radius. This is an *informational* entropy production—the cost of retrodiction from infinity to radius r—not a Clausius entropy. The Clausius entropy production for a photon in Tolman thermal equilibrium vanishes, since $T(r)\sqrt{-g_{00}} = T_\infty$ [?]. The quantity $\Sigma_{\rm grav}$ persists even in equilibrium: it measures the informational cost of climbing the gravitational potential.

Three independent arguments converge on Eq. (**??**), all sharing the mathematical core $\Sigma_{\rm grav} = -\ln(-g_{00})$:

(a) Modular flow [known].—In algebraic QFT on Schwarzschild, the quantum relative entropy (QRE) between the vacuum and a coherent excitation is computable via modular theory [?]. The Tolman blueshift and energy redshift together give a fractional QRE loss $(D_{\rm in} - D_{\rm out})/D_{\rm in} = r_{\rm s}/r$, exactly.

(b) Gravitational Landauer principle [known].—The Tolman-modified cost of erasing one bit at radius r exceeds the cost at infinity by $1/\sqrt{-g_{00}}$ [?]. The dimensionless excess erasure cost yields $\Sigma_{\rm grav} = -\ln(-g_{00})$, which equals $r_{\rm s}/r$ for the exponential metric and $-\ln(1 - r_{\rm s}/r) \approx r_{\rm s}/r + O((r_{\rm s}/r)^2)$ for Schwarzschild.

(c) Quantum channel [known].—Modeling gravitational redshift as a pure-loss bosonic channel with intensity transmissivity $\eta = -g_{00}$, the QRE decrease is $\Sigma = -\ln\eta = r_{\rm s}/r$ [?]. Petz recovery for this channel type has been experimentally demonstrated [?].

No route constructs a canonical gravitational CPTP map with explicit Kraus operators [?]. We bypass this: any consistent classical-gravity–quantum-matter coupling is irreversible [?], guaranteeing the existence of an irreversible channel; route (a) fixes the coefficient; route (c) provides a concrete existence proof. The modular channel of Trejo-Calderón [?] is the most promising candidate for a first-principles construction.

Substituting Eq. (**??**) into the Petz bound (**??**) gives

$F_{\rm bound} = e^{-\Sigma_{\rm grav}/2} = e^{\Phi/c^2}$. Comparing with Eq. (**??**):

$$\boxed{\frac{c_{\rm eff}}{c} \;=\; e^{-\Sigma_{\rm grav}/2} \;=\; F_{\rm bound}\,.} \tag{4}$$

This is the central result [new synthesis]: the coordinate speed of light *is* the Petz recovery fidelity bound. Equivalently, the gravitational refractive index is the inverse fidelity: $n_{\rm grav} = c/c_{\rm eff} = 1/F_{\rm bound} = e^{+\Sigma_{\rm grav}/2}$.

The physical picture is transparent. An alternative route to this identification uses quantum hypothesis testing [**?**]: $\Sigma_{\rm grav} = D(\rho(r_1)\|\rho(r_2))$, where $\rho(r)$ is the quantum state of a probe at radius r. This definition bypasses the need for an explicit CPTP map and gives $\Sigma_{\rm grav} = -\ln(-g_{00})$ directly from the Kubo–Martin–Schwinger (KMS) state structure in curved spacetime.

A photon climbing out of a gravitational potential loses information—its quantum state degrades due to interaction with the gravitational channel. The recovery fidelity $F_{\rm bound}$ quantifies how well the original state can be retrodicted. At the would-be event horizon, $F_{\rm bound} \to 0$ and $c_{\rm eff} \to 0$: complete information loss coincides with the vanishing of the coordinate speed of light. The event horizon *is* an information horizon, literally.

Table **??** verifies the identification across four orders of magnitude in gravitational compactness. The agreement is exact for the exponential metric and holds through $O(r_{\rm s}/r)$ for Schwarzschild; the first discrepancy is at $O((r_{\rm s}/r)^3)$, which is $\sim 10^{-18}$ in the solar system [**?**]—far below measurability.

CONSEQUENCES

Model-independent no-horizon theorem.—The Petz bound alone, before specifying $\Sigma_{\rm grav}$, has a consequence: if gravitational redshift is a quantum channel with *finite* entropy production, then $F \geq e^{-\Sigma/2} > 0$, so $\tau < 1$. An event horizon requires $F = 0$ (complete information loss), which demands $\Sigma \to \infty$—impossible for a finite-mass object at finite distance. The Petz bound forbids event horizons model-independently [**?**]; the specific form of $\Sigma_{\rm grav}$ is not needed.

The exponential metric.—The identity $\sqrt{-g_{00}} = e^{-\Sigma_{\rm grav}/2}$, taken as exact, determines the metric uniquely [**?** **?**]:

$$ds^2 = -e^{-r_{\rm s}/r}\,c^2dt^2 + e^{+r_{\rm s}/r}\big(dr^2 + r^2d\Omega^2\big)\,, \tag{5}$$

in isotropic coordinates. Since $e^{-r_{\rm s}/r} > 0$ for all $r > 0$, there is no event horizon: $\tau < 1$ everywhere. This metric has a long independent pedigree: Papapetrou [**?**], Yilmaz [**?**], Rosen's bimetric theory [**?**], and three scalar-field families within GR that all reduce to it when scalar charge equals gravitational mass [**?**]. Boonserm et al. classified it as a traversable wormhole with throat at $R_{\rm thr} = (e/2)\,r_{\rm s}$ [**?**]. It shares PPN parameters $\gamma = \beta = 1$ with Schwarzschild and passes all solar-system tests [**?**]. Our contribution is the information-theoretic motivation: the exponential metric is the geometry whose redshift factor *equals* the Petz recovery fidelity bound, $\sqrt{-g_{00}} = e^{-\Sigma/2}$, taken as an exact identity rather than a one-sided inequality. The spatial component $g_{rr} = e^{+r_{\rm s}/r}$ follows from the isometry condition $g_{00} \cdot g_{rr} = -1$ (Dicke's scalar refractive index [**?**]), which is an independent assumption—information theory constrains only g_{00}. Predictions are thus *robust* (no-horizon theorem, $n_{\rm grav} = 1/F_{\rm bound}$, decoherence bound—depending only on g_{00}) or *conditional* (shadow, QNM, echoes—depending on the full metric).

Caveat on saturation.—The JRSWW bound [**?**] is structurally *strict* for $\Sigma > 0$: the proof proceeds via the Golden–Thompson inequality, which is an equality only when the relevant operators commute—precisely the $\Sigma = 0$ case [**?**]. No known quantum channel saturates the bound for $\Sigma > 0$. In weak fields, the gap is $O((r_{\rm s}/r)^2)$, far below current measurability. The exponential metric (**??**) should therefore be understood not as the unique consequence of saturation, but as the geometry defined by the *identity* $\sqrt{-g_{00}} = e^{-\Sigma_{\rm grav}/2}$, which holds independently of whether the Petz bound is tight. Whether nature selects this geometry (exponential metric) or admits a gap (a metric between exponential and Schwarzschild) is empirical. The exponential metric is *not* a vacuum solution of the Einstein equations; it requires a phantom scalar source [**?** **?**]. We motivate the *metric* from information theory, not field equations—an important distinction from the Yilmaz program [**?**], whose modified field equations face well-documented difficulties [**?**].

Gravitational decoherence.—Pikovski et al. [**?**] showed that gravitational time dilation universally dephases composite quantum systems in spatial superposition. In the τ framework [**?**], the resulting temporal asymmetry is $\tau_{\rm grav}(t) = 1 - e^{-N\Gamma t^2}$, where $\Gamma \propto (\Delta E)^2(\Delta\Phi)^2/(\hbar^2c^4)$. The dephasing rate is controlled by $\Delta\Phi/c^2 \sim \Sigma_{\rm grav}/2$: the gravitational entropy production governs both the static (refractive index) and dynamic (decoherence) aspects of gravitational information loss. Bonder et al. [**?**] noted that the Pikovski effect is frame-dependent; our framework is consistent with this, since

TABLE I. Numerical verification of the triple identification. Φ/c^2 is the dimensionless potential, $\Sigma_{\rm grav} = 2|\Phi|/c^2$, and $F_{\rm bound} = e^{-\Sigma_{\rm grav}/2} = c_{\rm eff}/c$.

Location	Φ/c^2	$\Sigma_{\rm grav}$	$F_{\rm bound}$	$c_{\rm eff}/c$
Earth	-7×10^{-10}	1.4×10^{-9}	$1-7\times10^{-10}$	$1-7\times10^{-10}$
Sun	-2.1×10^{-6}	4.2×10^{-6}	$1-2.1\times10^{-6}$	$1-2.1\times10^{-6}$
NS	-0.2	0.4	0.819	0.819
Horizon	$-\infty$	∞	0	0

τ depends on the observer's channel, not on an absolute decoherence rate.

RELATION TO PRIOR WORK

The logic here is complementary to the Jacobson program [?]. Jacobson derives *field equations* from thermodynamic entropy applied to local Rindler horizons: $\delta Q = T\delta S \to$ Einstein equations $\to$ Schwarzschild. We derive the *metric component* directly from information recoverability: $F_{\rm bound} = e^{-\Sigma_{\rm grav}/2} \to g_{00} \to$ exponential metric. Both programs share the premise that geometry is information-theoretic; they differ in what is derived first (field equations vs. metric) and consequently arrive at different strong-field predictions. The Dorau–Much result [?]—deriving the semiclassical Einstein equations from quantum relative entropy via modular theory—suggests that the two programs share a common informational foundation and may ultimately converge.

In holographic settings, the Petz map reconstructs bulk quantum fields from boundary data [?], with fidelity $F_{\rm bulk} \geq e^{-\Delta S_{\rm gen}/2}$, where $\Delta S_{\rm gen}$ is the change in generalized entropy. Our identification (**??**) extends this structure beyond AdS/CFT: the bulk metric *itself* encodes the recovery fidelity, with $\sqrt{-g_{00}}$ playing the role of $F_{\rm bound}$. The Crooks fluctuation theorem has recently been established in curved spacetimes [?], confirming that the Crooks–Petz connection of Paper I [?] extends to general-relativistic settings—entropy production in curved spacetime is observer-dependent and quantified by $\Sigma_{\rm grav}$.

Galley, Giacomini, and Selby [?] proved that *any* consistent coupling between classical gravity and quantum matter is fundamentally irreversible ($\Sigma > 0$), providing model-independent support for the premise that gravitational processes produce entropy. Bai, Buscemi, and Scarani [?] independently constructed a fully quantum entropy production using Bayesian retrodiction—mathematically parallel to the τ framework but without gravitational application. De Vuyst et al. [?] showed via quantum reference frames that gravitational entropy is observer-dependent, providing algebraic justification for the coordinate dependence of $\Sigma_{\rm grav}$.

TESTABLE PREDICTIONS

The exponential and Schwarzschild metrics agree through second post-Newtonian order but diverge in strong fields, yielding three testable predictions ordered by near-term feasibility.

(1) Shadow size.—The exponential photon sphere at $r_{\rm ph} = r_{\rm s}$ (isotropic coordinates) yields a critical impact parameter $b_{\rm crit} = 2e\,m \approx 5.44\,m$ versus Schwarzschild's $3\sqrt{3}\,m \approx 5.20\,m$ ($m = GM/c^2$): a 4.6% larger shadow. For M87*, this gives 41.5 μas versus 39.7 μas; the EHT measurement of 42±3 μas [?] is consistent with both but closer to the exponential value. Next-generation EHT, targeting sub-μas precision, can distinguish the two.

(2) Quasinormal modes.—In the eikonal limit, $f_{\rm QNM}^{\rm exp}/f_{\rm QNM}^{\rm Schw} = 3\sqrt{3}/(2e) \approx 0.956$, a $\sim 4.4\%$ downward shift. This estimate should be treated with caution: the exponential photon sphere coincides with the wormhole throat, where the effective potential differs qualitatively from Schwarzschild. WKB calculations for a generalized exponential wormhole find a -7.8% shift [?]. Current LIGO ringdown analyses constrain QNM deviations at the $\sim$ 5–10% level; next-generation detectors (Einstein Telescope, Cosmic Explorer) will reach $\sim 1\%$ precision.

(3) Gravitational-wave echoes.—The most distinctive signature. The exponential metric's traversable wormhole structure produces echoes—repeated, damped pulses following the initial ringdown—with time delay

$$\Delta t_{\rm echo} \approx \frac{4.2\,r_{\rm s}}{c}\,, \tag{6}$$

giving $\sim$ 2.5 ms for a $60\,M_\odot$ remnant at $\sim$ 400 Hz, squarely in the LIGO band. This is $\sim 90\times$ shorter than Planck-wall models [?], because the exponential metric has no horizon and its tortoise coordinate $r^* = \int e^{r_{\rm s}/r}\,dr$ remains bounded. Current echo searches use $\sim$ 200 ms templates; dedicated short-delay searches are needed to test this prediction.

SCOPE AND LIMITATIONS

The identification (**??**) *can*: unify quantum error correction, thermodynamic irreversibility, and gravitational physics under a single parameter τ; provide universal recovery bounds that hold independently of the specific protocol; connect the Hayden–Preskill decoder—which *is* the Petz map [?]—to gravitational physics; and provide a precise, computable definition of the gravitational arrow of time: $\tau > 0$ whenever $\Sigma_{\rm grav} > 0$, with $\tau = 0$ if and only if spacetime is flat.

It *cannot*: explain dark matter ($\Sigma_{\rm grav} \sim 10^{-6}$ at galactic scales, far too small); Bianconi's gravity-from-entropy framework [?] yields Schwarzschild with $O(r^{-4})$ corrections in vacuum [?], *not* the exponential metric; our information-theoretic derivation via $\Sigma_{\rm grav} = -\ln(-g_{00})$ is independent of that program. It cannot explain the cosmological constant or dark energy; derive the full metric tensor g_{ab} from τ (the relation $g_{00} \to \Sigma_{\rm grav} \to \tau$ is one-directional; τ constrains only g_{00}, not the spatial components); or extend to general dynamical spacetimes without modification. A promising extension uses the Kodama vector [?]—the unique divergence-free vector in spherically symmetric spacetimes—giving $\tau_K(R) = 1 - \sqrt{1 - H^2R^2}$ in FRW, tracking apparent (not event)

horizons; the deeper foundation is modular flow via crossed-product algebras [**? ?**].

CONCLUSION

The coordinate speed of light, the informational entropy production cost, and the Petz recovery fidelity bound are the same number: $c_{\text{eff}}/c = e^{-\Sigma_{\text{grav}}/2} = F_{\text{bound}}$. This is a new connection, not a new theory—each piece is known; the synthesis is new. The exponential metric follows from the identity $\sqrt{-g_{00}} = e^{-\Sigma_{\text{grav}}/2}$, with testable strong-field deviations from Schwarzschild. The no-horizon theorem holds model-independently: for any finite Σ_{grav}, $\tau < 1$ and complete information loss cannot occur—regardless of whether the Petz bound is tight or strict.

The deepest implication is that the metric tensor does not merely *resemble* an information-theoretic quantity—$\sqrt{-g_{00}}$ *is* the recovery fidelity bound. Whether nature saturates this bound is a question that the strong-field observational program of the coming decade can answer.

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[1] A. Einstein, Ann. Phys. (Leipzig) **35**, 898 (1911).
[2] R. H. Dicke, Rev. Mod. Phys. **29**, 363 (1957).
[3] D. Petz, Q. J. Math. **39**, 97 (1988).
[4] M. Junge, R. Renner, D. Sutter, M. M. Wilde, and A. Winter, Ann. Henri Poincaré **19**, 2955 (2018).
[5] A. J. Parzygnat and F. Buscemi, Quantum **7**, 1013 (2023).
[6] S.-K. Huang, "The Arrow of Time from Petz Recovery," companion paper (2026).
[7] R. C. Tolman, Phys. Rev. **35**, 904 (1930).
[8] L. Dorau and A. Much, Phys. Rev. Lett. (2025); arXiv:2510.24491.
[9] L. Herrera, Entropy **22**, 340 (2020).
[10] M. Ahmadi, D. E. Bruschi, C. Sabin, G. Adesso, and I. Fuentes, Sci. Rep. **4**, 4996 (2014).
[11] B. Bertotti, L. Iess, and P. Tortora, Nature **425**, 374 (2003).
[12] A. Papapetrou, Z. Phys. **139**, 518 (1954).
[13] H. Yilmaz, Phys. Rev. **111**, 1417 (1958).
[14] N. Rosen, Gen. Relativ. Gravit. **4**, 435 (1973).
[15] M. A. Makukov and E. G. Mychelkin, Found. Phys. **50**, 1346 (2020).
[16] P. Boonserm, T. Ngampitipan, A. Simpson, and M. Visser, Phys. Rev. D **98**, 084048 (2018).
[17] C. W. Misner, arXiv:gr-qc/9504050 (1995).
[18] I. Pikovski, M. Zych, F. Costa, and Č. Brukner, Nat. Phys. **11**, 668 (2015).
[19] Y. Bonder, E. Okon, and D. Sudarsky, Phys. Rev. D **92**, 124050 (2015).
[20] T. Jacobson, Phys. Rev. Lett. **75**, 1260 (1995).
[21] J. Cotler, P. Hayden, G. Penington, G. Salton, B. Swingle, and M. Walter, Phys. Rev. X **9**, 031011 (2019).
[22] M. L. W. Basso, J. Maziero, and L. C. Celeri, Phys. Rev. Lett. **134**, 050406 (2025).
[23] Event Horizon Telescope Collaboration, Astrophys. J. Lett. **875**, L1 (2019).
[24] S. Nath and D. Sarma, arXiv:2401.03738 (2024).
[25] V. Cardoso, E. Franzato, and P. Pani, Phys. Rev. Lett. **116**, 171101 (2016).
[26] M. Hayashi and T. Yamasaki, Nat. Phys. (2025).
[27] Y. B. Bai, F. Buscemi, and V. Scarani, "Quantum stochastic entropy production from Bayesian retrodiction," (2024); arXiv:2412.12489.
[28] T. D. Galley, F. Giacomini, and J. O. E. Selby, "Any consistent coupling between classical gravity and quantum matter is fundamentally irreversible," Quantum **7**, 1142 (2023).
[29] J. De Vuyst, S. Eccles, P. A. Höhn, and J. Kirklin, "Gravitational entropy is observer-dependent," (2025); arXiv:2412.15502.
[30] Z. Chen, Y. Song, and V. Scarani, "Petz recovery from subsystems in bosonic quantum error correction," (2025); arXiv:2511.05941.
[31] B. Thattarampilly, Y. Zheng, and K. K. Kakkat, "Spherically symmetric black hole in gravity from entropy," (2026); arXiv:2602.13694.
[32] G. Bianconi, Phys. Rev. D **111**, 066001 (2025); arXiv:2408.14391.
[33] J. A. Trejo-Calderón, arXiv:2504.20457 (2025).
[34] M. Kasprzak and E. Tjoa, arXiv:2408.00518 (2024).
[35] H. Kodama, Prog. Theor. Phys. **63**, 1217 (1980).
[36] E. Witten, J. High Energy Phys. **2022**(10), 008 (2022).
[37] V. Chandrasekaran, R. Longo, G. Penington, and E. Witten, J. High Energy Phys. **2023**(02), 082 (2023).